Biodiversity

Biodiversity: definition, levels and importance

Biodiversity is a combination of two words – biological and diversity, meaning diversity of life forms.

Bio = Life;

Diversity = Variety;

The 1992 Earth Summit in Rio de Janeiro defined biodiversity as:

The variability among living organisms from all sources, including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part: this includes diversity within species, between species and of ecosystems.

Biodiversity is the variety of life on Earth and the essential interdependence of all living things among themselves and with their environment. Biodiversity is an important factor for the successful functioning of the ecosystem.

Current estimates of no. of species on earth:

- 10-14 million
- Till date the scientists have identified more than 2 million species. Tens of millions -- remain unknown
- ~1 million are insects
- 99.9% (5 billion) of species extinct since the beginning of life on earth

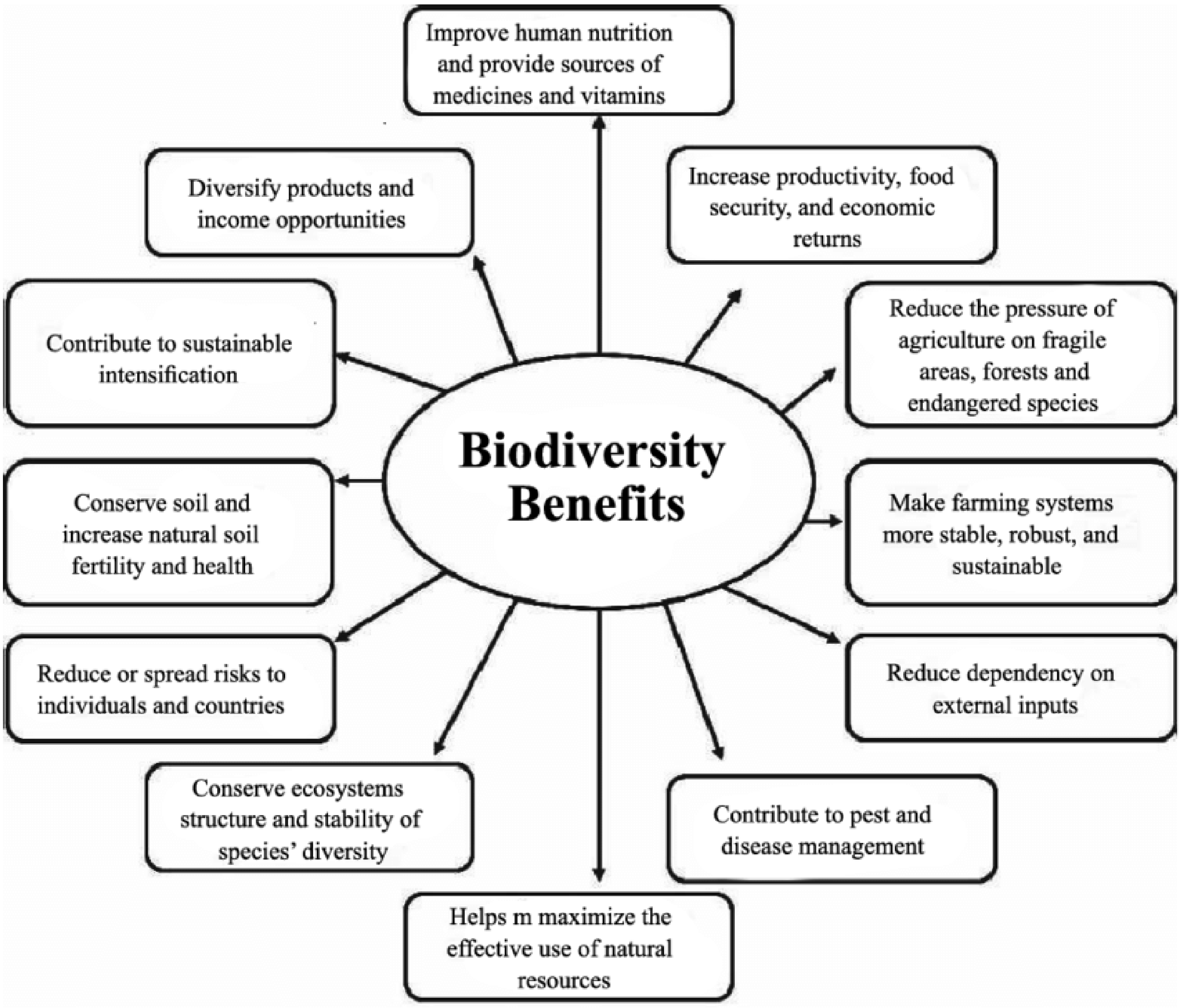
What is biodiversity and why is it important?

Biodiversity is generally defined as the number and variability of all the life forms pertaining to plants, animals and micro-organisms and the ecological complex they inhabit.

The two documented benefits of biodiversity are:

I. Consumptive and productive uses - grains, vegetables, fruits, plants, medicines, timber, oils, forest products, milk products, eggs, the list of items on this account is endless.
II. Non consumptive benefit where we have biodiversity's role in providing raw materials for biotechnology, regulation of water and other nutrient cycles, regulation of climatic conditions, carbon fixation etc.

The economic value of biodiversity is also of great benefit. "Each species is of potential value to humans. So are healthy ecosystems. The global collection of genes, species, habitats and ecosystems is a resource that provides for human needs now, and is essential for human survival in the future. Human depend on other species for all of their food and for many medicines and industrial products.



There are three levels of diversity 1) Genetic diversity, 2) Species diversity, 3) Ecosystem diversity

1. Genetic Diversity:

Genetic diversity is the "fundamental currency of diversity" that is responsible for variation. Genetic diversity is defined as genetic variability present within the species. It is the ability of an organism to adapt to changes in the local environment. They adapt by possession of different alleles suitable to the environment.

E.g., Different breeds of dogs, different varieties of roses, wheat, rice, mangoes, etc

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Genetic diversity is very important. It will ensure the survival and adaptability of the species during unfavorable survival conditions in the environment such as disease, or climate change

2. Species Diversity:

Species diversity is a major component of biodiversity and tends to increase the sustainability of some ecosystems. It is the most visible component of biodiversity as implied by the word 'species' which literally means outward or visible form.

How we will define the species diversity: Species diversity is the number of different species that are represented in a given community.

Or

Species diversity is defined as the number of different species present in an ecosystem and relative abundance of each of those species.

Diversity is greatest when all the species present are equally abundant in the area. There are two constituents of species diversity: i.e 1) Species richness, 2) Species evenness

Species richness:

The number of different species present in an ecosystem is species richness. Tropical areas have greater species richness as the environment is helpful for a large number of species.

Species evenness:

Species evenness is a description of the distribution of abundance across the species in a community. Species evenness is highest when all species in a sample have the same abundance. Evenness approaches zero as relative abundances vary.

It is possible in an ecosystem to have high species richness, but low species evenness.

Example:

In a forest, there may have a large number of different species (high species richness) but have only a few members of each species (low species evenness)

In a forest, there may be only a few plant species (low species richness) but a large number of each species (high species evenness)

Species richness increases with increasing explored area.

Why the species diversity is very important: Greater species diversity ensures sustainability in an ecosystem. Since each species is intertwined intricately uniquely with the ecosystem, each performing a unique role, extinction of even one species can have countless ripple effects on the entire ecosystem.

3. Ecosystem Diversity:

Ecosystem Diversity can be defined as the variety of different habitats, communities and ecological processes. A biological community is defined by the species that occupy a particular area and the interactions between those species. Groups of organisms and their non-living environment, and the interactions between them, form functional dynamic and complex units that are termed ecosystems. These systems help maintain life processes vital for organisms to survive on earth.

Species are not evenly distributed around the globe. Some ecosystems such as tropical rain forests and coral reefs are very complex and host a large number of species. Other ecosystems such as deserts and arctic regions have less biodiversity but are equally important.

Variations in food webs, nutrient cycles, trophic structure etc, this diversity has developed along with evolution

Eg: Tropical rainforests, deserts, ponds, oceans etc.

Prairies, Ponds, and tropical rain forests are all ecosystems. Each one is different, with its own set of species living in it.

How ecosystem diversity is very Importance: Biodiversity is the variety of life in an area that is determined by the number of different species in that area.

- Biodiversity increases the stability of an ecosystem and contributes to the health of the biosphere.
- Variations in ecosystems in a region, and its overall impact on human existence and environment

E.g: deserts, forests, grasslands, wetlands, oceans

- Greater diversity ensures sustainability and ecosystems capable of withstanding environmental stresses like floods, draughts, pests etc.
- Ecosystem diversity ensures availability of oxygen by photosynthesis
- In an aquatic environment, water purification is carried out by the various plant species
- Greater variety of plants, means a greater variety of crops

An ecosystem having higher diversity means the number of species and interactions between them which constitute the food web, is large. In such a situation, the elimination of one species would have little effect on ecosystem balance. In sharp contrast, the number of species in the food web of a simple ecosystem is small. So, loss of any one species has far more serious repercussions for the integrity of the ecosystem itself.

Values of biodiversity:

It has two different values. i. e intrinsic and utilitarian values

Intrinsic Value = Something that has value in and of itself

Utilitarian Value = It is useful to others

Biodiversity is the most precious gift of nature. We all know that all the organisms in an ecosystem are interlinked and interdependent. The importance of biodiversity in the life of all the organisms including humans is huge.

We all are getting benefits from biodiversity mainly in two ways.

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Firstly, biodiversity is directly used as a source for food, fibre, fuel and other extractable resources.

Secondly, biodiversity plays an important role in ecosystem processes providing the regulating, cultural and supporting services.

For example, vegetation cover protects the soil from erosion by binding soil particles and minimizing the effects of water runoff. Similarly, cultivation of crops is to a large extent dependent on the availability of pollinating insects.

Biodiversity has a fundamental value to humans because we are so dependent on it for our cultural, economic, and environmental well-being.

In the field of medicine alone, approximately 50% of current prescription medicines are derived from or modelled on natural substances. The health and diversity of ecosystems can have a significant effect on the overall stability of nearby communities.

Some of the major values of biodiversity are as follows:

Direct values:

Direct use values are for those goods that are ensured directly e.g. food and timber. Maintaining a wide range of components of biological diversity can be of direct use, especially in the fields of agriculture, medicine and industry. Direct use can involve the use of forests, wetlands or other ecosystems for timber extraction, collection of non-timber products, fishing, etc. Direct use values could be due to extractive use where resources are extracted and consumed, or due to non-extractive use when there is no extraction or removal of the resource that is used (e.g. bird watching, scientific research in an ecosystem).

Generally, it divided into two categories-

1. Consumptive use value

2. Productive Use Value

Indirect values:

1. Cultural and Social Value

2. Ecosystem Services

3. Economic Value

4. Ethical and Moral Value

5. Aesthetic Value.

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Direct values:

1. Consumptive use value:

These are direct use values where the biodiversity products are consumed or harvested directly. E.g.: fuel, food, drugs, fibre etc.

Humans use at least 40,000 species of plants and animals on a daily basis. Many people around the world still depend on wild species for most of their needs like food, shelter and clothing. The tribal people are completely dependent on the forests for their daily needs.

2. Productive Use Value

These are commercially usable values where the product is marketed and sold, often resulting in the exploitation of rich biodiversity.

This is assigned to products that are commercially harvested and marketed. Almost all the present date agricultural crops have originated from wild varieties. Biodiversity represents the original stock from which new varieties are being developed. The biotechnologists continuously use the wild species of plants for developing new, better yielding and disease resistant varieties.

Indirect values:

1. Cultural and Social Value:

Social value of biodiversity refers to its religious and cultural importance.

Certain customs and religious practices utilize plants for their rituals, and worship them as well. It revolves around utilization of either plants and/or animals for either rituals or are worshipped

Example: Trees like Peepal, Banyan and Tulsi are still worshipped. Ladies offering water to Tulsi daily is considered good and there are festivals when ladies tie sacred threads around Peepal and Banyan trees and pray for the welfare of their families.

Flowers and tulsi leaves are offered during poojas

Animals such as cows, snakes and other animals are worshipped in different religions

2. Ecosystem Services:

These services also support human needs and activities such as intensely managed production ecosystems.

Ecosystem service includes:

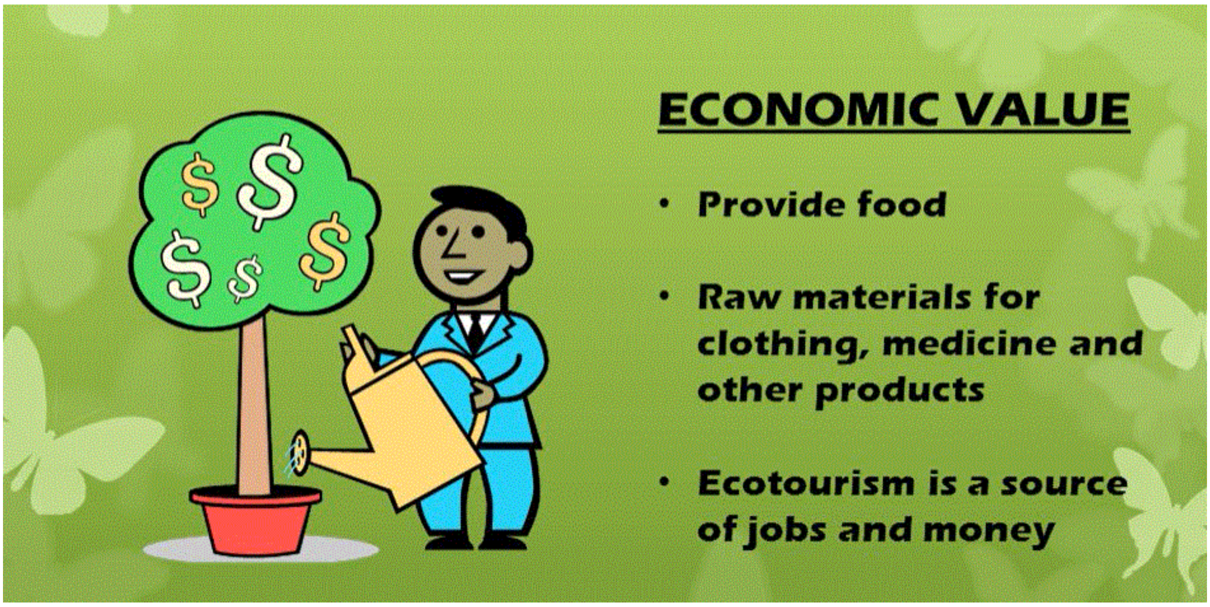
1. The production of oxygen by land-based plants and marine algae.

1. The provision of native species and genes used in industry research and development, for instance, in traditional breeding and biotechnology applications in agriculture, forestry, horticulture, mariculture, pharmacy, chemicals production and bioremediation;

2. Pollination of agricultural crops, forest trees and native flowering plants by native insects, birds and other creatures;
3. Maintenance of habitats for native plants and animals; and Maintenance of habitats that are attractive to humans for recreation, tourism and cultural activities and that has spiritual importance.

3. Economic Value:

The economic potential of biodiversity is immense in terms of food~~, fodder~~, medicinal, ethical and social values. Biodiversity forms the major resource for different industries, which govern the world economy.



The salient features regarding the economical potential of biodiversity are given below:

1. The major fuel sources of the world including wood and fossil fuels have their origin due to biodiversity.

2. It is the source of food for all animals and humans.

3. Many important chemicals have their origin from the diverse flora and fauna, used in various industries.

4. Ethical and Moral Value:

It is based on the principle of ‘live and let others live’. Ethical values related to biodiversity conservation are based on the importance of protecting all forms of life. All forms of life have the right to exist on earth. Man is only a small part of the Earth’s great family of species.

Morality and ethics teach us to preserve all forms of life and not to harm any organism unnecessarily. Some people take pleasure in the hunting of animals. People also sometimes degrade and pollute the environment by their unethical actions.

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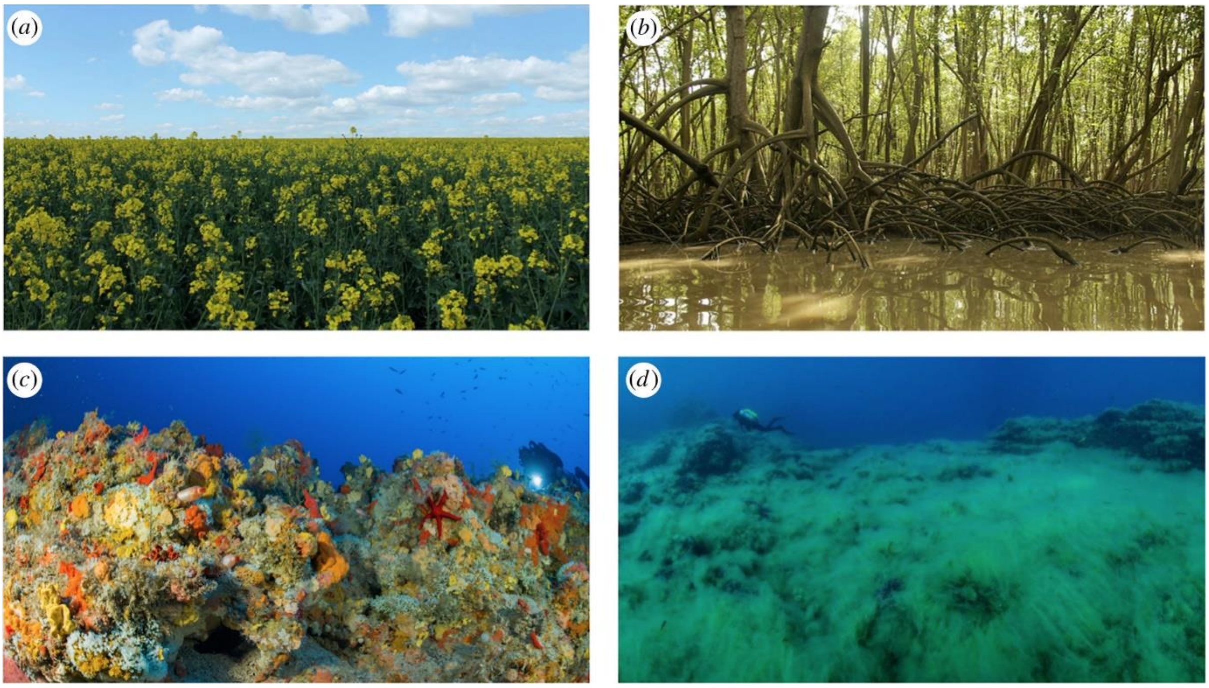
5. Aesthetic Value:

The beauty of our planet is because of biodiversity, which otherwise would have resembled other barren planets dotted around the universe. Biological diversity adds to the quality of life and provides some of the most beautiful aspects of our existence. Biodiversity is responsible for the beauty of a landscape.



People go far off places to enjoy the natural surroundings and wildlife. This type of tourism is referred to as eco-tourism, which has now become a major source of income in many countries. In many societies, the diversity of flora and fauna has become a part of the traditions and culture of the region and has added to the aesthetic values of the place.

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"There is enough for everyone's need but not for anyone's greed"

-Mahatma Gandhi